Technical report: EEG electrode impedance under VR headsets

BrainAccess MINI (8 channels) with HTC Vive and Meta Quest

Date: 2026-11-06
Prepared by: EEG-VR-stats project
Data: 1 participant, session S001, 3 recordings (no VR, HTC Vive, Meta Quest)

1 Summary

- **VR headsets did not raise impedance overall.** Median impedance across the 8 electrodes was 87 kΩ without VR, 83 kΩ with HTC Vive and 83 kΩ with Meta Quest. Differences were electrode-specific (e.g. O1/O2 lower and C3 higher under both headsets).
- **Impedance decreases after fitting and needs time to settle.** In the no-VR, HTC and Meta Quest recordings, 5, 5 and 5 of 8 electrodes, respectively, dropped by more than 10% between the first and last 10 s, mostly in steps. Electrodes that started high dropped the most (up to 1119 → 149 kΩ, F4, no VR). Several electrodes were still changing after 2 minutes.
- **Short impedance spikes occur** (up to 8288 kΩ with HTC, 4357 kΩ with Meta Quest). They last at most a few seconds.
- **Recommended smoothing: 5-s rolling median.** It removes the spikes but keeps real changes. The window length has almost no effect on the overall level or trend; longer windows (15–30 s) hide genuine short changes.

2 Setup

Device	BrainAccess MINI, dry electrodes, impedance mode
Electrodes	F3, F4, C3, C4, P3, P4, O1, O2
Sampling rate	250 Hz
Conditions (in order)	no VR (154 s), HTC Vive (156 s), Meta Quest (168 s)
Participant	1 (sub-1), single session

The VR headset was worn over the EEG headset. Recordings were exported from the acquisition software (SQLite) to CSV; dates, device serial numbers and computer timestamps were removed. No data were excluded or interpolated.

3 Processing

- Each electrode's impedance was smoothed with a centred rolling median of 1, 5, 15, 30 s. The median ignores isolated spikes.
- **Initial / final impedance:** median of the first / last 10 s.
- **Settling time:** last moment the 5-s median was more than max(10%, 10 kΩ) away from the final value.
- **Trend:** robust (Theil–Sen) slope of the impedance over time, in kΩ/min.

4 Results

4.1 Impedance per condition

Figure 1 shows impedance over time for every electrode, and Table 1 gives the numbers. Electrode medians ranged from 51 (P4) to 152 kΩ (F4) without VR, 14 (O2) to 226 kΩ (P3) with HTC Vive and 27 (O1) to 225 kΩ (C3) with Meta Quest. There was no consistent shift between conditions (Friedman test across electrodes, $p = 0.325$); with one participant this is indicative only.

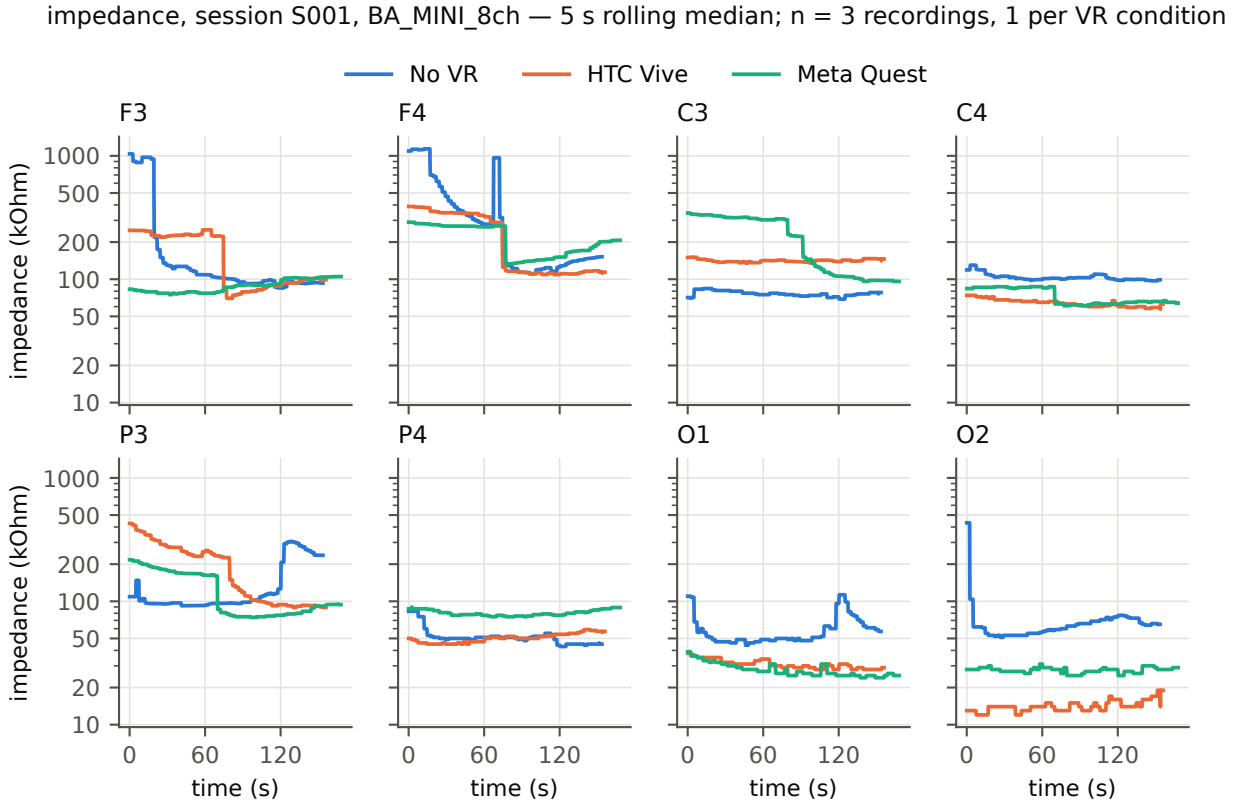


Figure 1: Impedance (kΩ, log scale) per electrode, 5-s rolling median, one recording per condition (250 Hz; $n = 38406, 39000, 42000$ samples).

Table 1: Impedance per electrode (kΩ): median [interquartile range], initial → final value, and settling time (s).

	No VR			HTC Vive			Meta Quest		
Ch.	median [IQR]	init. → final	settle (s)	median [IQR]	init. → final	settle (s)	median [IQR]	init. → final	settle (s)
F3	103 [94–127]	902 → 93	70	102 [87–227]	248 → 101	120	87 [79–100]	81 → 105	120
F4	152 [128–431]	1119 → 149	127	119 [113–344]	385 → 115	75	206 [151–270]	284 → 206	149
C3	76 [74–79]	81 → 78	0	141 [139–143]	150 → 146	0	225 [105–316]	337 → 96	118
C4	102 [100–106]	120 → 98	111	64 [61–67]	74 → 59	22	66 [63–86]	85 → 65	70
P3	98 [95–126]	109 → 237	142	226 [94–267]	378 → 92	99	92 [78–170]	212 → 94	144
P4	51 [48–52]	83 → 45	15	50 [47–54]	48 → 57	48	79 [77–84]	87 → 89	125
O1	51 [48–61]	68 → 60	137	31 [29–33]	36 → 28	0	27 [25–30]	36 → 25	7
O2	63 [55–69]	64 → 65	127	14 [13–15]	13 → 17	0	28 [26–29]	28 → 28	0

4.2 Changes during the recording

- Impedance fell by more than 10% in 5 (no VR), 5 (HTC) and 5 (Meta Quest) of 8 electrodes and rose in 1, 2 and 1.

- Largest drops: F4 1119 → 149 kΩ (no VR), P3 378 → 92 kΩ (HTC), C3 337 → 96 kΩ (Meta Quest).
- Median trend: −3.5 (no VR), −4.1 (HTC) and −5.9 kΩ/min (Meta Quest).
- Median settling time: 119 s (no VR), 35 s (HTC), 119 s (Meta Quest). Only 2, 5 and 2 electrodes, respectively, were stable from 60 s onwards. Late changes also occurred (e.g. P3 rose after about 2 minutes without VR).

4.3 Choice of smoothing window

Rolling medians of all four lengths agree for most of the time and differ only around spikes and short changes (Fig. 2 and Appendix A). Table 2 and Fig. 3 summarise the effect:

- The trend estimate barely depends on the window. With 1-s vs. 30-s windows it was −3.3 vs. −3.3 (no VR), −4.1 vs. −4.5 (HTC) and −5.8 vs. −7.9 kΩ/min (Meta Quest; at 30 s based on only 5 points).
- A 1-s window removed the HTC spike (8288 kΩ, F3) but not the Meta Quest excursion (4357 kΩ, F4); a 5-s window removed both.
- 15–30 s windows also remove genuine short changes, e.g. the brief rise of F4 at about 65 s without VR, which is visible only at 1 and 5 s.

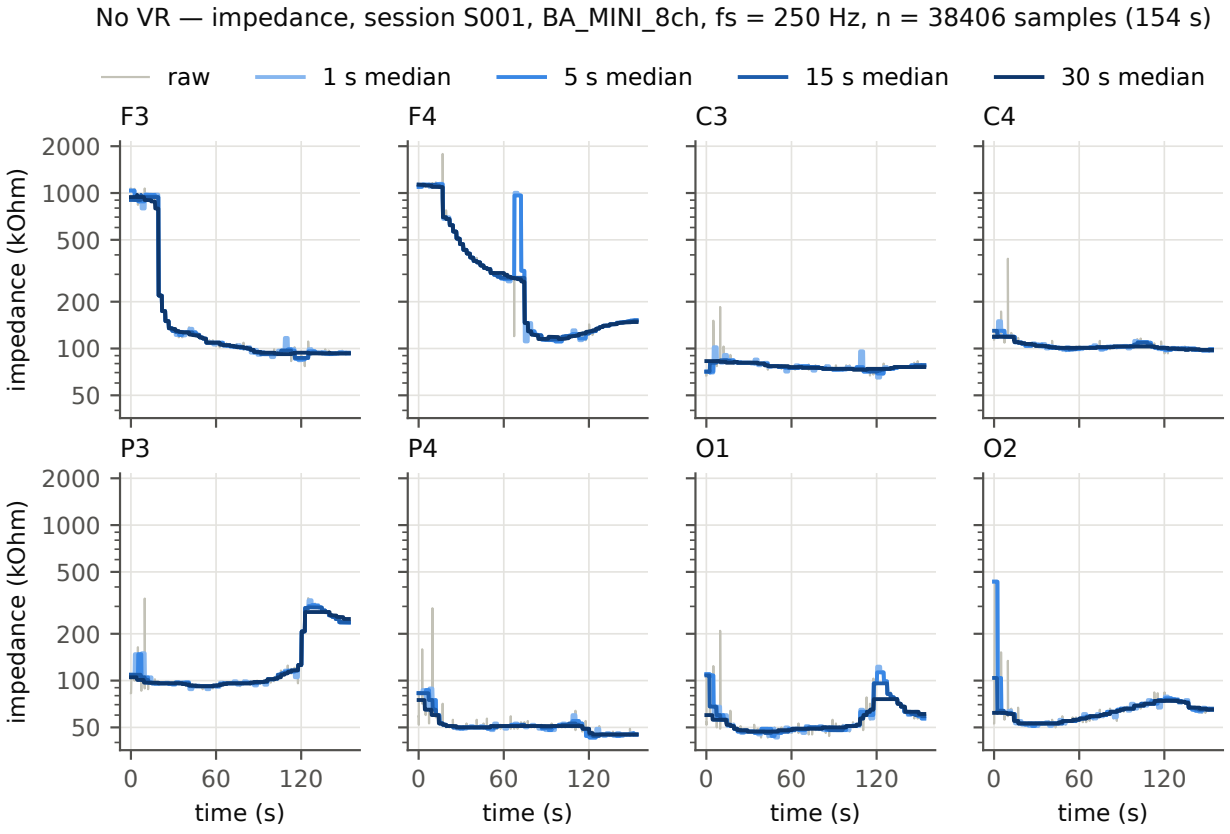


Figure 2: No-VR recording: raw impedance (grey) and rolling medians of 1, 5, 15, 30 s (light to dark blue). Log scale.

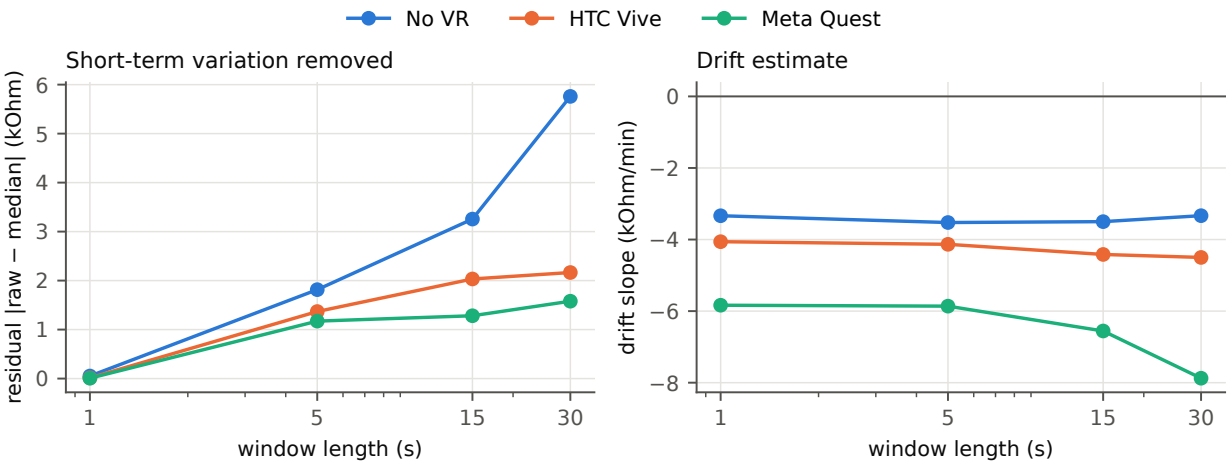


Figure 3: Effect of window length. Left: average difference between raw and smoothed impedance (what the window removes). Right: trend estimate. Median over 8 electrodes per recording.

Table 2: Effect of window length: average removed variation (kΩ) and trend (kΩ/min), median over 8 electrodes.

	No VR		HTC Vive		Meta Quest	
Window (s)	resid. (kΩ)	slope (kΩ/min)	resid. (kΩ)	slope (kΩ/min)	resid. (kΩ)	slope (kΩ/min)
1	0.1	−3.3	0.0	−4.1	0.0	−5.8
5	1.8	−3.5	1.4	−4.1	1.2	−5.9
15	3.3	−3.5	2.0	−4.4	1.3	−6.6
30	5.8	−3.3	2.2	−4.5	1.6	−7.9

5 Recommendations

1. Display or log impedance with a **5-s rolling median**; use longer windows only for reporting slow trends.
2. Allow a **settling period** after fitting the EEG headset and again after putting on the VR headset, and check impedance again before recording. In these data, 2.5–3 minutes were not enough for all electrodes.
3. Check **frontal electrodes (F3, F4)** in particular: they had the highest initial impedance and largest drops, also without a VR headset, so this reflects fitting rather than the VR headset.

6 Limitations

One participant and one recording per condition, recorded in a fixed order (no VR, HTC, Meta Quest); differences between conditions may therefore also reflect time since fitting or re-fitting of the headset. Results should be confirmed with more participants and a balanced order of conditions.

A Window comparison for the VR conditions

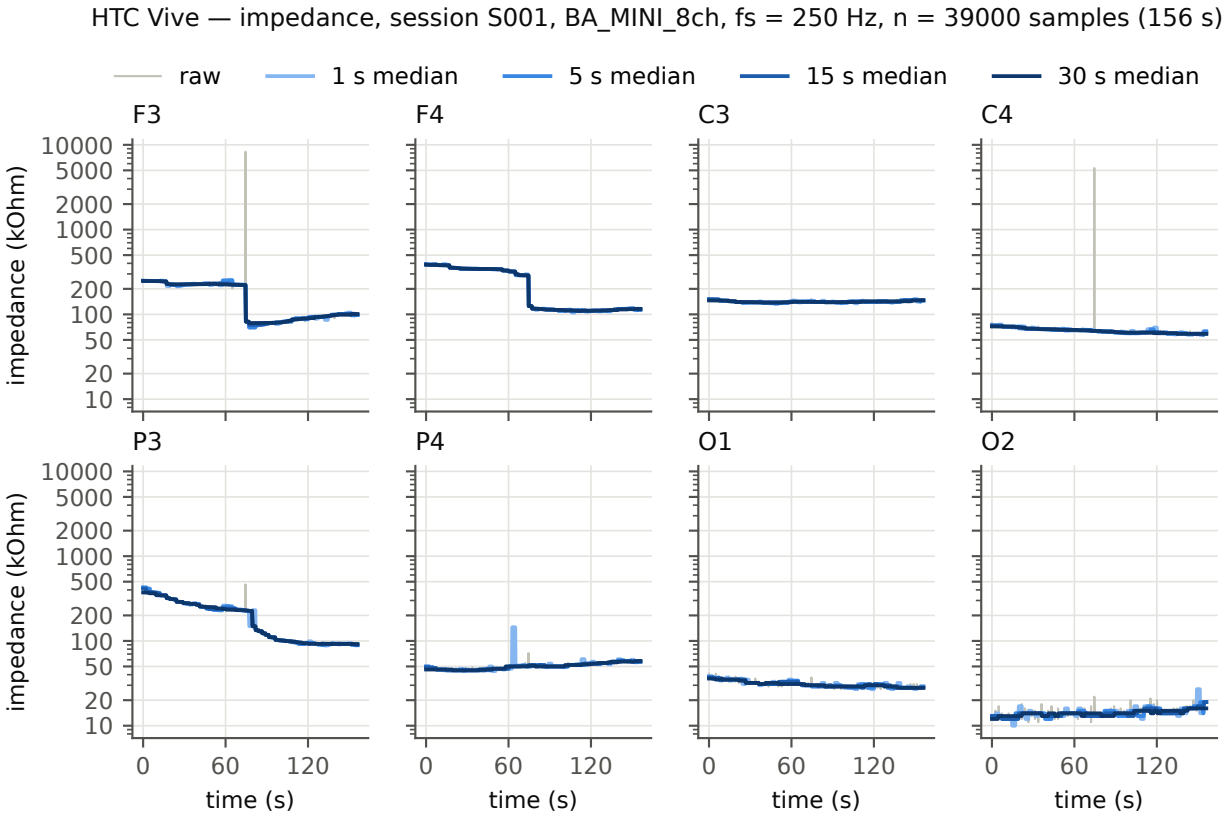


Figure 4: HTC Vive recording; as Fig. 2.

B Data and reproducibility

Generated by `processing/impedance/impedance_over_time.ipynb` from `datafiles/impedance/impedance-S001-*-BA_MINI_8ch.csv`; parameters and source checksums are in `outputs/impedance/impedance_provenance_impedance-S001-BA_MINI_8ch.json`.

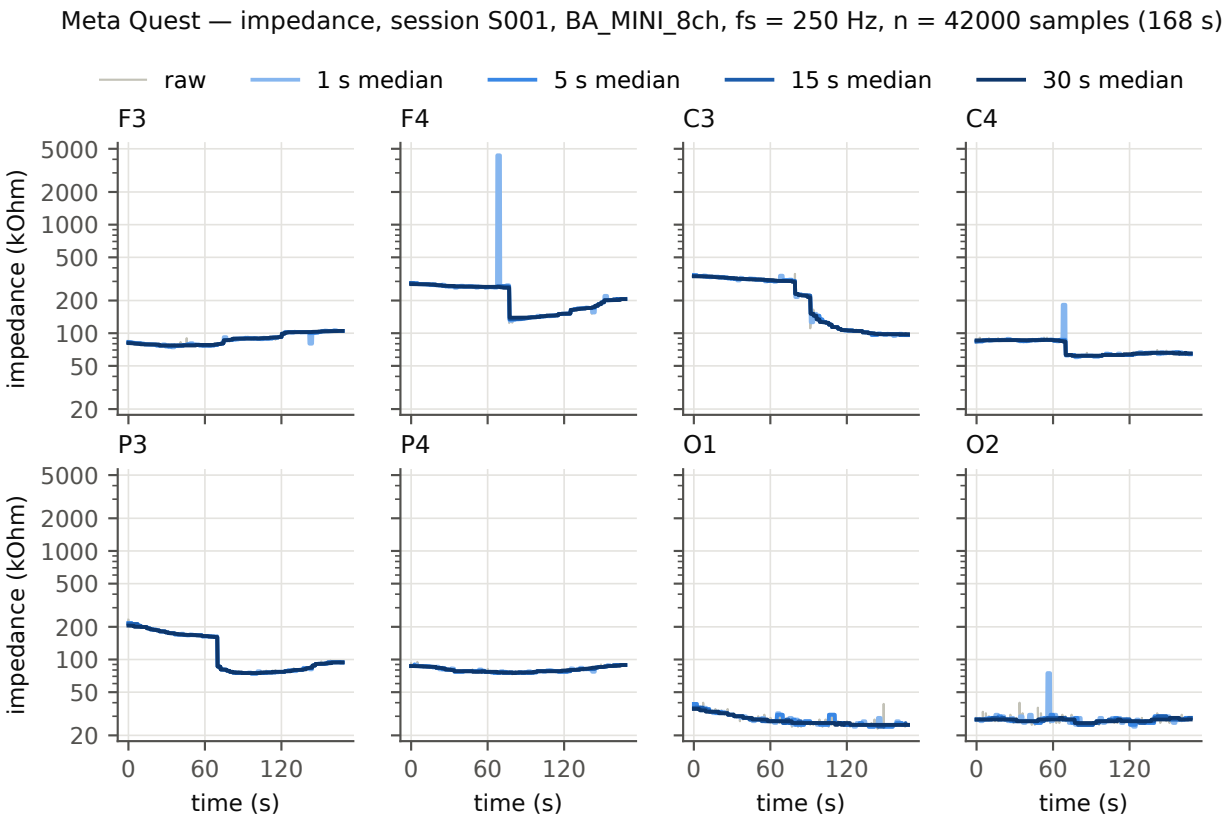


Figure 5: Meta Quest recording; as Fig. 2.